CHEME 480 Lab 2 Reflection Mark A. Heien

1. What was the goal of Lab 1? How did it relate to class?
The goal of Lab 1 was to collect temperature-response data from the TCLab and fit that data to a dynamic model. It related to class because I was connecting the differential equation and first-order response we discuss in lecture to real experimental data.

2. How does the objective of Lab 1 relate to the objective of Lab 2? What do they do the same, and what are we doing differently in Lab 2?
Both Lab 1 and Lab 2 had me compare a mathematical model to real TCLab temperature data. The difference is that Lab 1 focused on a simpler dynamic model, while Lab 2 had me compare different heat-transfer models and examine how the sensor and actuator affect the measured signal.

3. Why do we have to run the same code again in the Run TCLab section as we did in Lab 1?
I had to run the code again because I still needed fresh temperature data from the device. The model comparison in Lab 2 depends on having actual measured data from the TCLab, so I had to repeat the experiment and save the new response.

4. In this lab we used odeint. What is an alternative function we have learned in prior courses that we could have used instead to do the same thing? What do both of these functions do?
An alternative I could have used is ode45. Both odeint and ode45 numerically solve ordinary differential equations, meaning they calculate how the system changes over time according to the model equations.

5. What was the difference between the two model systems we tried to fit the data to today? Did the difference between the two change the model significantly? Why or why not?
The first model included only convective heat loss, while the second included both convective and radiative heat losses. In this case, adding radiation did not change the model much because the radiation term was small over the temperature range we tested; both models gave very similar predictions.

6. What is the point of Lab 2 Part B: Sensors and Actuators? What did we learn about devices from doing this?
The point of Part B was to show that a real device does not measure temperature as a perfect continuous value. I learned that the sensor converts temperature into voltage, the Arduino converts that voltage into discrete digital levels, and that quantization and noise can make measured data differ from the exact model.